

Ranges and
Definitions

3 Steps to ABG
Interpretation

Apply steps 1&2:

Acidemia

Normal pH

Alkalemia

Compensation

CASES

Click these buttons to
REVEAL & HIDE
answers and graphics

Ranges and Definitions

3 Steps to ABG Interpretation

Apply steps 1&2:

Acidemia

Normal pH

Alkalemia

Compensation

CASES

Normal PCO_2 is 40 ± 4
Respiratory acidosis - $\text{pCO}_2 > 44$
Respiratory alkalosis - $\text{pCO}_2 < 36$

Normal HCO_3 is 24 ± 2
Metabolic acidosis - $\text{HCO}_3 < 22$
Metabolic alkalosis - $\text{HCO}_3 > 26$

pH

PCO_2

PO_2

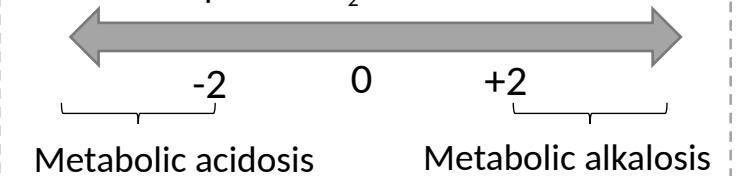
HCO_3

Normal pH is 7.40 ± 0.05
Alkalemia - $\text{pH} > 7.45$
Acidemia - $\text{pH} < 7.35$

Normal PO_2 on an ABG is > 80 .
Check out or talk on hypoxemia [here](#).

Base excess

Some labs will report a "base excess". This is the # of H^+ ions required to restore normal pH if PCO_2 is normal.



- "-emia" refers to the pH
- "-osis" refers to a metabolic or respiratory process that would lead to a derangement in pH if unchecked

Ranges and
Definitions

3 Steps to ABG
Interpretation

Apply steps 1&2:

Acidemia

Normal pH

Alkalemia

Compensation

CASES

Use the menu on the left-hand side of the page to navigate to each page

Step 1: Acid-Base status?

Acidemia pH
<7.35

Normal pH
7.35 – 7.45

Alkalemia
pH > 7.45

Primary process?

Metabolic? Respiratory? Or both?

Step 2: Anion gap metabolic acidosis?

An anion gap metabolic acidosis can be present even with a primary respiratory process or primary metabolic alkalosis.

If yes, is there a concurrent metabolic process? (Delta-delta)

Step 3: Assess for compensation

Ranges and
Definitions

Step 1

3 Steps to ABG
Interpretation

Apply steps 1&2:

Acidemia

Normal pH

Alkalemia

Compensation

CASES

Metabolic
 $\text{HCO}_3^- < 22$

Acidemia $\text{pH} < 7.35$

Respiratory
 $\text{pCO}_2 > 44$

Acidemia pH < 7.35

Ranges and Definitions

Step 1

3 Steps to ABG Interpretation

Apply steps 1&2:

Step 2

Acidemia

Normal pH

Alkalemia

Compensation

CASES

Metabolic
 $\text{HCO}_3^- < 22$

↑ Anion gap (AG)?

No

Yes

The Anion Gap

Cations

Anions

Na^+

HCO_3^-

Cl^-

K^+

albumin

PO_4^{3-} , others

Measured AG = $(\text{Na}) - (\text{HCO}_3 + \text{Cl})$.

Compare the calculated and expected AG. Normally, the expected AG is 8-12. But because it's predominantly albumin, the expected AG is ↓ when albumin is ↓.

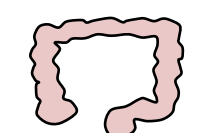
Expected AG = albumin x 2.5

If the calculated AG > expected AG, there is an AG.

Respiratory
 $\text{pCO}_2 > 44$

Urinary AG

NeGUTive



GI loss of HCO_3^-

Diarrhea
Ileostomy
Enteric fistulas

Positive



Renal loss of HCO_3^-

↑ Cl^- (excess NS)
RTA
Early renal failure
↓ aldosterone
Acetazolamide

$\Delta\Delta$ (delta-delta) or "residual HCO_3^- "
Is there another metabolic process?

<1

1

>2

AGMA +

Pure

AGMA +

Delta-delta ($\Delta\Delta$)

$$\frac{\text{Measured AG} - \text{Expected AG}}{\text{Measured HCO}_3^- - 24} = \frac{\Delta\text{AG}}{\Delta\text{HCO}_3^-}$$

- $\Delta\Delta < 1 \rightarrow$ concurrent NAGMA
- $\Delta\Delta = 1 \rightarrow$ pure AGMA
- $\Delta\Delta = 1-2 \rightarrow$ pure AGMA OR concurrent metabolic alkalosis
- $\Delta\Delta > 2 \rightarrow$ concurrent metabolic alkalosis

"Residual" $\text{HCO}_3^- = \Delta\text{AG} + \text{measured HCO}_3^-$

Residual process is left if the AGMA is accounted for?
 $\text{HCO}_3^- < 22 \rightarrow$ concurrent NAGMA
 $\text{HCO}_3^- 22-26 \rightarrow$ pure AG
 $\text{HCO}_3^- > 26 \rightarrow$ concurrent metabolic alkalosis

The Urine AG

Cations

Anions

Na^+

Cl^-

K^+

Other anions
(HCO_3^- , PO_4^{3-})

UAG

Urine AG = $\text{U}_{\text{Na}} + \text{U}_{\text{K}} - \text{U}_{\text{Cl}}$

The UAG measures urine NH_4^+ , aka urine excretion of H^+ .

- GI loss of $\text{HCO}_3^- \rightarrow$ compensatory ↑ in urine $\text{NH}_4^+ \rightarrow$ UAG is negGUTive.
- In RTAs, the secretion of NH_4^+ is impaired.

The UAG is not accurate in patients with underlying kidney disease, ketonuria. In these cases, the **urine osmolar gap** should be used instead.

Ranges and
Definitions

Step 1

3 Steps for ABG
Interpretation

Apply steps 1&2:

Acidemia

Normal pH

Alkalemia

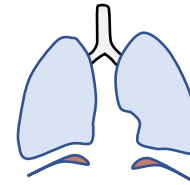
Compensation

CASES

Metabolic
 $\text{HCO}_3^- < 22$

Acidemia $\text{pH} < 7.35$

Respiratory
 $\text{pCO}_2 > 44$



↓ CO_2
elimination

Airway obstruction
Asthma, COPD
Intrinsic lung disease
Neuromuscular weakness
↓ of RR (opioids, benzos)

↑ CO_2
production

Sepsis
Overfeeding
Hyperthyroidism
↑ exercise

Ranges and Definitions

3 Steps to ABG Interpretation

Apply steps 1&2:

Acidemia

Normal pH

Alkalemia

Compensation

CASES

Step 1

Step 2

Normal pH

Look for metabolic (HCO_3^-) or respiratory (PCO_2) derangements that might suggest mixed acid-base disorder

↑ Anion gap (AG)?

Yes

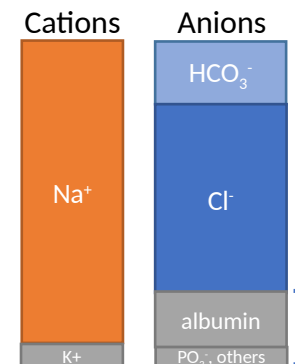
$\Delta\Delta$ (delta-delta) or "residual HCO_3^- "
Is there another metabolic process?

Mixed AGMA + NAGMA

Pure AGMA

Mixed AGMA + metabolic alkalosis

The Anion Gap



Measured AG = $(\text{Na}) - (\text{HCO}_3 + \text{Cl})$.

Compare the calculated and expected AG. Normally, the expected AG is 8-12. But because it's predominantly albumin, the expected AG is ↓ when albumin is ↓.

Expected AG = albumin x 2.5

If the calculated AG > expected AG, there is an AG.

Delta-delta ($\Delta\Delta$)

$$\frac{\text{Measured AG} - \text{Expected AG}}{\text{Measured HCO}_3 - 24} = \frac{\Delta\text{AG}}{\Delta\text{HCO}_3}$$

- $\Delta\Delta < 1 \rightarrow$ concurrent NAGMA
- $\Delta\Delta 1 \rightarrow$ pure AGMA
- $\Delta\Delta 1-2 \rightarrow$ pure AGMA OR concurrent metabolic alkalosis
- $\Delta\Delta > 2 \rightarrow$ concurrent metabolic alkalosis

"Residual" HCO_3^- = $\Delta\text{AG} + \text{measured HCO}_3^-$

AKA, what metabolic process is left if the AGMA is accounted for?

- "Residual" $\text{HCO}_3^- < 22 \rightarrow$ concurrent NAGMA
- "Residual" $\text{HCO}_3^- 22-26 \rightarrow$ pure AG
- "Residual" $\text{HCO}_3^- > 26 \rightarrow$ concurrent metabolic alkalosis

Ranges and Definitions

3 Steps to ABG Interpretation

Apply steps 1&2:

Acidemia

Normal pH

Alkalemia

Compensation

CASES

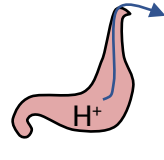
Step 1

Alkalemia pH > 7.45

Metabolic
 $\text{HCO}_3^- > 26$

Respiratory
 $\text{PCO}_2 < 36$

GI loss



Vomiting
NGT

Renal loss



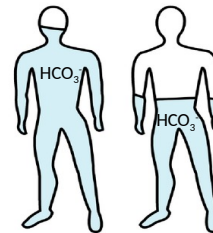
Thiazides
Loop diuretics
↑ aldosterone
Genetic causes
(Barter's, Gitelman's)

Other causes

Milk alkali syndrome

Hypokalemia

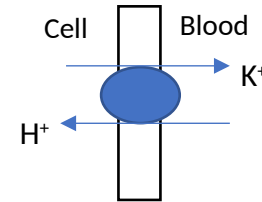
Contraction



Response to IVF?

The differential for metabolic alkalosis can be narrowed by response to IVF (essentially, Cl^- infusion). Only GI loss, diuretic use and contraction alkalosis are responsive to IVF. Nonresponse to IVF suggests an alternative etiology.

Anxiety, pain
CNS disease (tumor, infection, trauma, CVA)
Fever, sepsis
Drugs (salicylates)
Liver disease
Pregnancy
PE



Ranges and Definitions

3 Steps to ABG Interpretation

Apply steps 1&2:

Acidemia

Normal pH

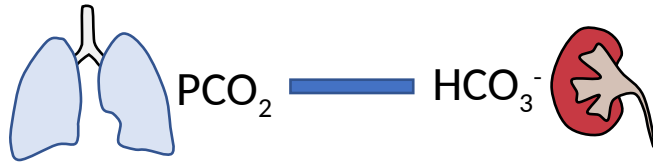
Alkalemia

Compensation

CASES

COMPENSATION

Step 3: COMPENSATION



The kidney and lungs help regulate pH balance. The HCO_3^- and PCO_2 move in the same direction.

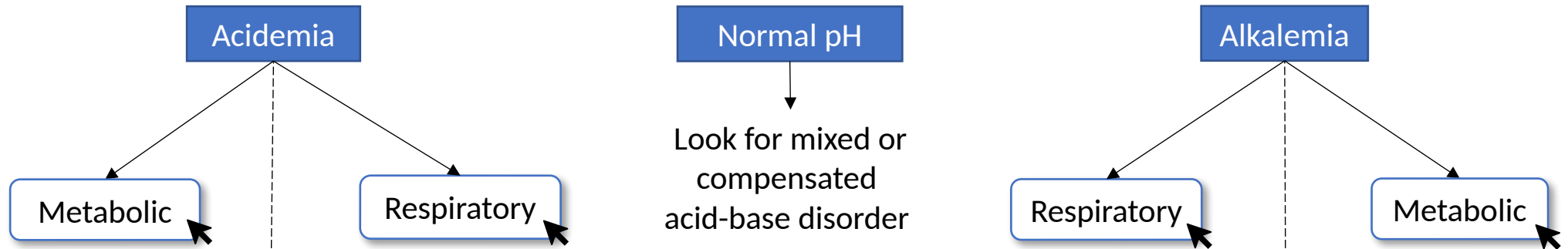
- When HCO_3^- is $\downarrow$ (metabolic alkalosis), the body will eliminate CO_2 ($\downarrow$) to compensate.
- When PCO_2 is $\uparrow$ (respiratory acidosis), the body will retain HCO_3^- ($\uparrow$) to compensate.

Respiratory (PCO_2) compensation occurs rapidly (minutes).

Metabolic (HCO_3^-) compensation occurs more slowly (2-3 days).

Compensation will never overcompensate or fully correct pH to normal (7.40).

PRIMARY PROCESS



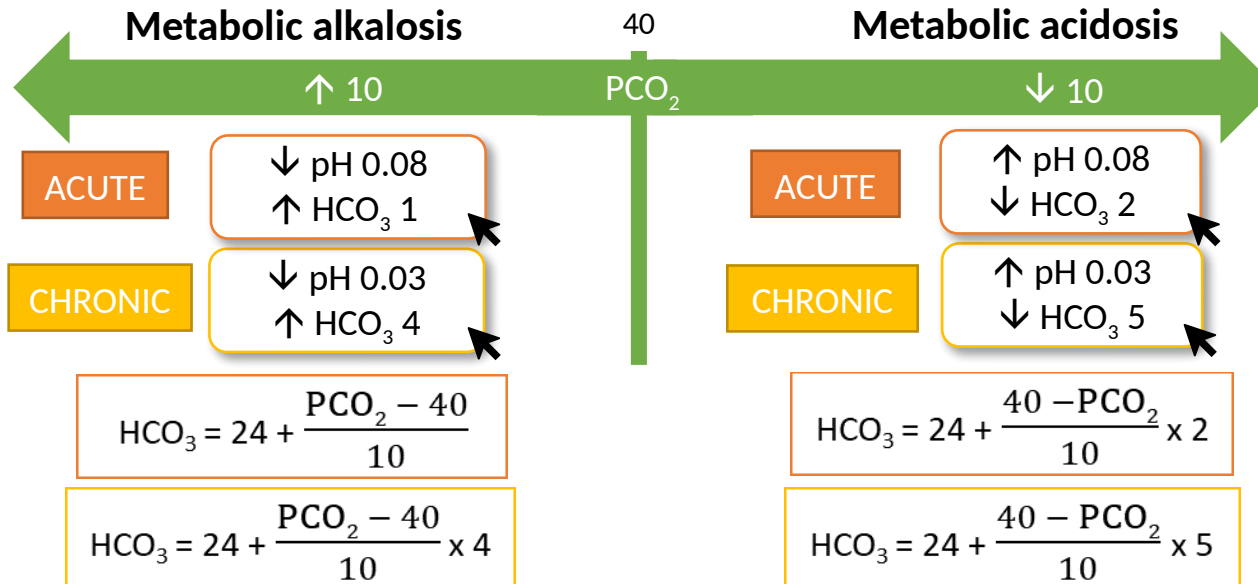
Respiratory alkalosis

1. Winter's formula
 $\text{PCO}_2 = (1.5 \times \text{HCO}_3^-) + 8$

2. Thumb rule
 PCO_2 should be $\approx$ to last digits of pH
.32/30/-/18



Metabolic alkalosis



Respiratory acidosis

$$\text{PCO}_2 = (0.7 \times \text{HCO}_3^-) + 21$$

Ranges and Definitions

3 Steps to ABG Interpretation

Apply steps 1&2:

Acidemia

Normal pH

Alkalemia

Compensation

CASES

Basic

1-3

4-5

Click each shaded box to reveal the answer.

↓pH / ↓PCO₂ / PO₂ / ↓HCO₃

Acidemia

Metabolic acidosis

↓ PCO₂

↓pH / ↑PCO₂ / PO₂ / ↓HCO₃

Acidemia

Concurrent metabolic and respiratory acidosis

-

↓pH / ↑PCO₂ / PO₂ / ↑HCO₃

Acidemia

Respiratory acidosis

↑ HCO₃

↑pH / ↑PCO₂ / PO₂ / ↑HCO₃

Alkalemia

Metabolic alkalosis

↑ PCO₂

↑pH / ↑PCO₂ / PO₂ / ↓HCO₃

Alkalemia

Respiratory alkalosis

↓ HCO₃

↑pH / ↓PCO₂ / PO₂ / ↑HCO₃

Alkalemia

Concurrent metabolic and respiratory alkalosis

-

Ranges and Definitions

3 Steps to ABG Interpretation

Apply steps 1&2:

Acidemia

Normal pH

Alkalemia

Compensation

CASES

Basic

1-3

4-5

Case 1

34F is found down. In the ED, her SaO₂ was 89% on RA. Pinpoint pupils on exam. ABG: 7.25/60/60/26, base excess of 1.

136	100		Albumin 4.0
	26		

Acidemia. pH of 7.25 is low.

Respiratory acidosis. PCO₂ is ↑ (nml 36-44) driving the acidemia. HCO₃ is at the upper range of normal.

There is no concurrent AGMA.
(Na) - (HCO₃ + Cl) = 136 - (100 + 26)
AG = 10 (normal 8-12)

No ΔΔ needed.

None. HCO₃ is nml. No base excess. For acute PCO₂ ↑ 20 → ↑ HCO₃ by 2 (26).

A-a difference is 15, normal for her age.

Acute respiratory acidosis and hypoxemia from hypoventilation likely from narcotic overdose.

Case 2

45M with IBD presents with abdominal pain and multiple loose bowel movements. ABG: 7.34/35/-/18.

127	100		Albumin 4.0
	18		

Acidemia. pH is borderline low.

Metabolic acidosis. Both PCO₂ and HCO₃ are ↓. Because the pH is borderline low, the metabolic acidosis is the 1° process.

There is no AGMA. This is a NAGMA.
(Na) - (HCO₃ + Cl) = 127 - (100 + 18)
AG = 9 (normal 8-12)

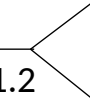
No ΔΔ needed. UAG would not be accurate given renal dysfunction.

Yes. Resp alkalosis. Expected PCO₂ by Winter's formula = (1.5 x HCO₃) + 8 = 35. The expected PCO₂ by the thumb rule is also 35.

Non-gap metabolic acidosis with compensatory respiratory alkalosis from diarrhea.

Case 3

65M with COPD presents with fever and cough. His SaO₂ is 90% on 30% FiO₂ by Venturi mask. ABG: 7.36/60/60/36.

142	97		Albumin 3.0
	32		

pH is low-normal.

Respiratory acidosis. Both PCO₂ and HCO₃ are ↑. The pH is low-nml, thus, respiratory acidosis is the 1° process.

There is no AGMA.
(Na) - (HCO₃ + Cl) = 142 - (97 + 36) = 9
Expected AG = 2.5 x albumin = 7.5.

No ΔΔ needed.

Yes. Met alkalosis. pH is nml so likely chronic. For chronic PCO₂ ↑ 20 → ↑ HCO₃ by 8 (32).

A-a difference is 72, ↑ for his age.

Chronic, compensated resp acidosis. High A-a gradient hypoxemia may be related to VQ mismatch or shunt.

Ranges and Definitions

3 Steps to ABG Interpretation

Apply steps 1&2:

Acidemia

Normal pH

Alkalemia

Compensation

CASES

Basic

1-3

4-5

Case 4

72 yo F with RYGB presents with 3 d of shortness of breath, abdominal pain, nausea and diarrhea. She received IV morphine for her pain and became somnolent. An ABG showed: 6.98/41/58/10.

146	121	18	98
	7	1.0	

Lipase 3290 Lactate 2.7
Albumin 2.1
Urine: + ketones

Acidemia. pH of 6.98 is low.

Metabolic acidosis (HCO_3 is $\downarrow$). PCO_2 is normal, but there is likely a concurrent primary respiratory process (see compensation).

There is an $\uparrow$ AG.
The AG = $146 - (121 + 7) = 18$.
The differential is MUDPILES.

There is a concurrent NAGMA (likely from diarrhea).
Using the "residual HCO_3 " method, the $\Delta\text{AG} = 18 - \text{expected AG}$. With an albumin of 2.1, the expected AG is 5-6. $\Delta\text{AG} = 12$.
Adding ΔAG to HCO_3 of 7 = 19, which is still a metabolic acidosis.

Using Winter's formula: Expected $\text{PCO}_2 = (1.5 \times \text{HCO}_3) + 8 = 19$.
Measured PCO_2 is 41, suggesting a concurrent respiratory acidosis.

There is AGMA from ketoacidosis and lactic acidosis, NAGMA likely from diarrhea, and a concurrent respiratory acidosis likely from narcotic associated hypoventilation.

Case 5

24 yo M with a history of depression is brought into the ED unresponsive. On exam, his RR is 34 breaths/min, SaO_2 99% on RA. ABG is 7.56/22/110/17.

137	99		
	18	1.2	

Albumin 4.0
Lactate 1.2
Urine: bland

Alkalemia. pH is high.

Respiratory alkalosis. PCO_2 is $\downarrow$.

There is an $\uparrow$ AG. AGMAs are never a compensatory process so there is a concurrent primary AGMA.
 $\text{AG} = 137 - (99 + 18) = 20$

This is a pure AGMA.
Using the "residual HCO_3 " method, the $\Delta\text{AG} = 20 - \text{expected AG (12)}$. $\Delta\text{AG} = 8$. If we add this to measured HCO_3 , the "residual HCO_3 " is 26, which is upper limit of normal.

None. There is a concurrent primary AGMA.

A primary respiratory alkalosis and AGMA is classic for salicylate poisoning.